Math 54 — Week 8

Solutions to the problem set

P1. **Idea.** Find eigenvalues and eigenvectors, then use $A^k = PD^kP^{-1}$.

1. $\operatorname{tr} A = 3$ and $\det A = 2 - 12 = -10$, so
$$\det(A - \lambda I) = \lambda^2 - 3\lambda - 10 = (\lambda - 5)(\lambda + 2),$$
and the eigenvalues are 5 and -2.
2. For $\lambda = 5$, the first row of $A - 5I$ is $(-3, 3)$, so $x_1 = x_2$: eigenvector $(1, 1)$. For $\lambda = -2$, the first row of $A + 2I$ is $(4, 3)$, so $4x_1 + 3x_2 = 0$: eigenvector $(-3, 4)$.
3. Two distinct eigenvalues, so A is diagonalizable (Theorem 6). With $\det P = 4 + 3 = 7$,
$$P = \begin{bmatrix} 1 & -3 \\ 1 & 4 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 0 \\ 0 & -2 \end{bmatrix}, \quad P^{-1} = \frac{1}{7}\begin{bmatrix} 4 & 3 \\ -1 & 1 \end{bmatrix}.$$
4. Then
$$A^k = PD^kP^{-1} = \frac{1}{7}\begin{bmatrix} 5^k & -3(-2)^k \\ 5^k & 4(-2)^k \end{bmatrix}\begin{bmatrix} 4 & 3 \\ -1 & 1 \end{bmatrix} = \frac{1}{7}\begin{bmatrix} 4\cdot 5^k + 3(-2)^k & 3\cdot 5^k - 3(-2)^k \\ 4\cdot 5^k - 4(-2)^k & 3\cdot 5^k + 4(-2)^k \end{bmatrix}.$$
5. Check with $k = 1$:
$$\frac{1}{7}\begin{bmatrix} 20 - 6 & 15 + 6 \\ 20 + 8 & 15 - 8 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} = A.$$

P2. **Idea.** The eigenvalues can be read off (triangular, or one expansion); the question is the dimension of the eigenspace for the repeated eigenvalue.

1. A is lower triangular with diagonal $2, 2, 3$. The matrix
$$A - 2I = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$
has rank 2, so $\dim \operatorname{Nul}(A - 2I) = 1$, less than the multiplicity 2. A is not diagonalizable (Theorem 7b).

P2, continued: the matrix B.

1. For B, expand $\det(B-\lambda I)$ down column 2, whose only nonzero entry is $5-\lambda$:
$$\det(B-\lambda I) = (5-\lambda)\det\begin{bmatrix}4-\lambda & -2\\ 0 & 5-\lambda\end{bmatrix} = (5-\lambda)^2(4-\lambda).$$
2. $\lambda = 5$: the matrix below has rank 1 (row 2 is -2 times row 1), so the eigenspace has dimension 2; its equation is $x_1 = -2x_3$ with x_2, x_3 free, giving the basis $(0,1,0)$, $(-2,0,1)$.
$$B-5I = \begin{bmatrix}-1 & 0 & -2\\ 2 & 0 & 4\\ 0 & 0 & 0\end{bmatrix}$$
3. $\lambda = 4$: rows 1 and 3 of the matrix below give $x_3 = 0$, and then row 2 gives $2x_1 + x_2 = 0$: eigenvector $(-1,2,0)$.
$$B-4I = \begin{bmatrix}0 & 0 & -2\\ 2 & 1 & 4\\ 0 & 0 & 1\end{bmatrix}$$
4. The dimensions add up to 3, so B is diagonalizable, and $BP = PD$ for
$$P = \begin{bmatrix}-1 & 0 & -2\\ 2 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}, \qquad D = \begin{bmatrix}4 & 0 & 0\\ 0 & 5 & 0\\ 0 & 0 & 5\end{bmatrix}.$$

P3. Idea. Count eigenspace dimensions: each simple eigenvalue gives exactly 1 (Theorem 7a), and the repeated one gives $4 - \operatorname{rank}(A-I)$ (rank theorem).

1. The eigenvalues 2 and 3 have multiplicity 1, so by Theorem 7a their eigenspaces have dimension 1.
2. If $\operatorname{rank}(A-I) = 3$, then $\dim \operatorname{Nul}(A-I) = 1$, and the total is $1+1+1 = 3 < 4$: not diagonalizable (Theorem 7b).
3. If $\operatorname{rank}(A-I) = 2$, then $\dim \operatorname{Nul}(A-I) = 2$, the total is 4, and A is diagonalizable.

P4. Idea. The columns of the matrix are the images of the $\mathcal{B}$ vectors, in $\mathcal{C}$-coordinates.

1. $T(1) = 1+t$ and $T(t) = t+t^2$, with $\mathcal{C}$-coordinates $(1,1,0)$ and $(0,1,1)$. So
$$M = \begin{bmatrix}1 & 0\\ 1 & 1\\ 0 & 1\end{bmatrix}.$$
2. $[\,3-2t\,]_{\mathcal{B}} = (3,-2)$, and
$$M\begin{bmatrix}3\\ -2\end{bmatrix} = \begin{bmatrix}3\\ 1\\ -2\end{bmatrix},$$
so $T(3-2t) = 3+t-2t^2$.
3. Check directly: $(t+1)(3-2t) = 3t-2t^2+3-2t = 3+t-2t^2$.

P5. Idea. Compute $A\mathbf{b}_1$ and $A\mathbf{b}_2$ and write them in $\mathcal{B}$-coordinates; these are the columns of the $\mathcal{B}$-matrix.

1. $A(1,1) = (3-1,\ 2) = (2,2) = 2(1,1)$ and $A(1,2) = (3-2,\ 2) = (1,2) = 1\cdot(1,2)$.
2. So the $\mathcal{B}$-coordinates of the images are $(2,0)$ and $(0,1)$, and the $\mathcal{B}$-matrix is $\operatorname{diag}(2,1)$.
3. It is diagonal because $(1,1)$ and $(1,2)$ are eigenvectors of A, with eigenvalues 2 and 1 (Theorem 8).

P6. Idea. Use one row of $A - \lambda I$ for the eigenvector, then Theorem 9 with $\lambda = a - bi$.

1. $\operatorname{tr} A = 6$ and $\det A = 8 + 2 = 10$, so $\det(A - \lambda I) = \lambda^2 - 6\lambda + 10$ and $\lambda = 3 \pm i$.
2. For $\lambda = 3 - i$, the first row of $A - \lambda I$ is $(-1 + i,\ -1)$. By the shortcut, an eigenvector is $\mathbf{v} = (-1,\ 1 - i)$. Check:
$$A\mathbf{v} = \begin{bmatrix} -2 - (1 - i) \\ -2 + 4(1 - i) \end{bmatrix} = \begin{bmatrix} -3 + i \\ 2 - 4i \end{bmatrix}, \qquad (3 - i)\mathbf{v} = \begin{bmatrix} -3 + i \\ (3 - i)(1 - i) \end{bmatrix} = \begin{bmatrix} -3 + i \\ 2 - 4i \end{bmatrix}.$$
3. $\operatorname{Re}\mathbf{v} = (-1, 1)$, $\operatorname{Im}\mathbf{v} = (0, -1)$, and $a = 3$, $b = 1$:
$$P = \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix}, \qquad C = \begin{bmatrix} 3 & -1 \\ 1 & 3 \end{bmatrix}, \qquad A = PCP^{-1}.$$
Check $AP = PC$: both have columns $(-3, 2)$ and $(1, -4)$.
4. The scale factor is $r = \sqrt{9 + 1} = \sqrt{10}$ (also $\sqrt{\det A}$), and the angle has $\cos\varphi = 3/\sqrt{10}$, $\sin\varphi = 1/\sqrt{10}$, so $\varphi \approx 18.4°$.

P7. Idea. Take inverses or transposes of $A = PDP^{-1}$; the middle factor stays diagonal.

1. Let $A = PDP^{-1}$. By the product rule, $\det A = \det P \det D \det P^{-1} = \det D = d_1 \cdots d_n$, which is not 0 since A is invertible. So every $d_i \neq 0$ and $D^{-1} = \operatorname{diag}(1/d_1, \ldots, 1/d_n)$ exists.
2. The inverse of a product is the product of the inverses in reverse order:
$$A^{-1} = (P^{-1})^{-1} D^{-1} P^{-1} = PD^{-1}P^{-1},$$
so A^{-1} is diagonalizable (with the same P).
3. Transposing reverses the order too:
$$A^T = (P^{-1})^T D^T P^T = QDQ^{-1} \quad \text{with } Q = (P^T)^{-1} = (P^{-1})^T,$$
because $D^T = D$ and $Q^{-1} = P^T$. So A^T is diagonalizable.

P8. Idea. Write $\mathbf{x}_0$ in the eigenvector basis; each piece is multiplied by its eigenvalue at every step.

1. Each row of A adds up to 1, so $A(1, 1) = (1, 1)$; and $A(-1, 1) = (-\frac{1}{2}, \frac{1}{2}) = \frac{1}{2}(-1, 1)$. The eigenvalues are 1 and $\frac{1}{2}$.
2. $\mathbf{x}_0 = (1, 0) = \frac{1}{2}(1, 1) - \frac{1}{2}(-1, 1)$.
3. So
$$\mathbf{x}_k = \frac{1}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{2}\left(\frac{1}{2}\right)^k \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} + \frac{1}{2^{k+1}} \\ \frac{1}{2} - \frac{1}{2^{k+1}} \end{bmatrix}.$$
For $k = 1$ this is $(\frac{3}{4}, \frac{1}{4}) = A\mathbf{x}_0$.
4. As $k \to \infty$, $(\frac{1}{2})^k \to 0$, so $\mathbf{x}_k \to (\frac{1}{2}, \frac{1}{2})$, the eigenvector for 1 whose entries add up to 1.